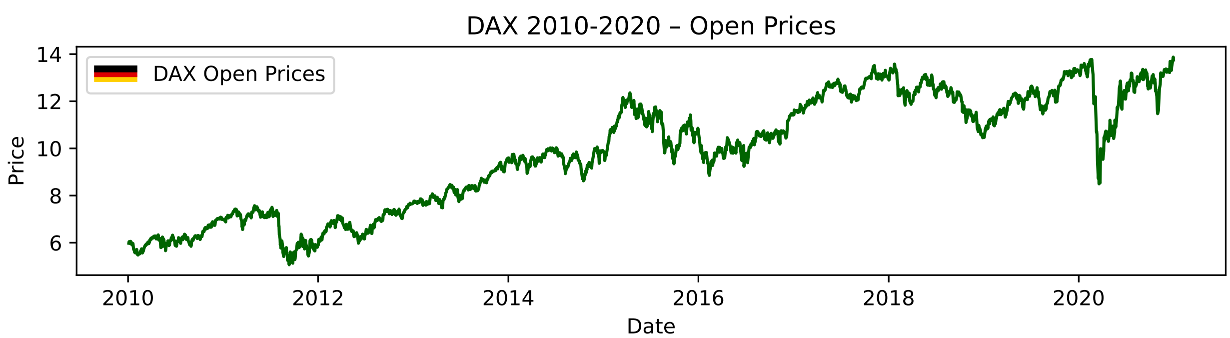
STATS 531 - Midterm Project

We look into the DAX Stock Index in Germany, and investigate how we can model the stock's opening prices, volatility, and other features. The DAX index tracks the performance of the 40 largest companies listed on the Regulated Market of the Frankfurt Stock Exchange (STOXX Ltd. 2026). They represent the diversified economy of Germany. The DAX is generally known as the German equity benchmark, surviving for over 30 years. What are the best ways to model changes in the stock's price and its returns? How do we model its volatility, and how accurate are our models?

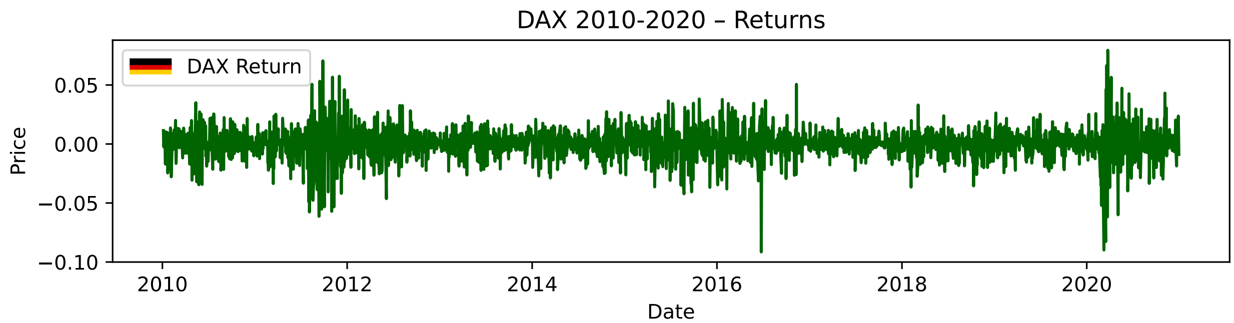
1 Exploratory Data Analysis

1.0.1 Introduction

We look at the daily stock price open for the DAX index for every day from 2010 to 2020 through a time series graph. The data above for stock open shows a roughly nonstationary data. The mean is not constant, and we have a large dip in 2020, likely associated with covid.



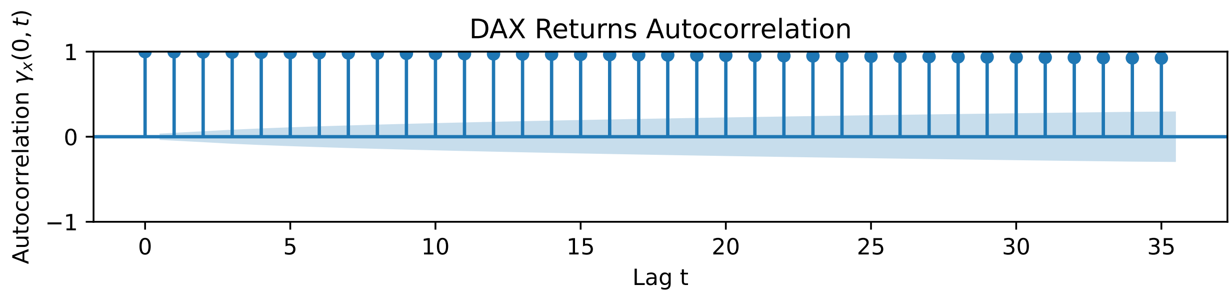
Now analyze volatility for the DAX stock, calculated as $y_n = log(z_n) - log(z_{n-1})$ (Ionides 2026).



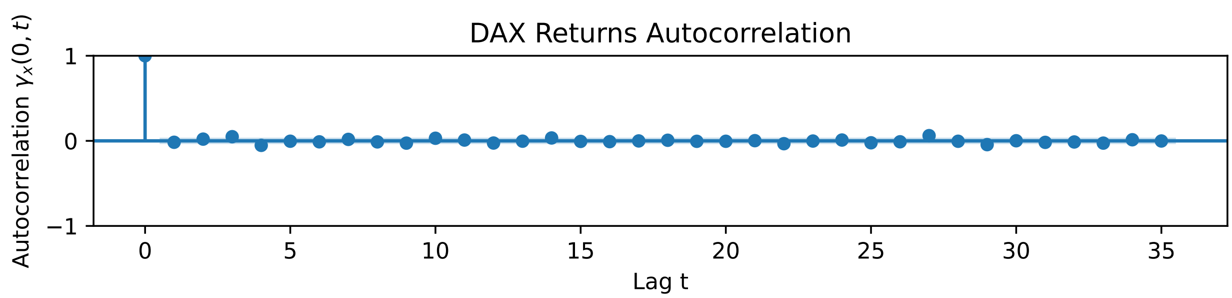
Through visual inspection, volatility is stationary with a more or less constant mean.

1.0.2 Autocorrelation

We look at the autocorrelation function below to see how much lags depend on each other. DAX opening prices are all highly correlated from the original value down until the 35th lag. Correlation then drops slowly. This tells us that a model that depends on its lags seems suitable.

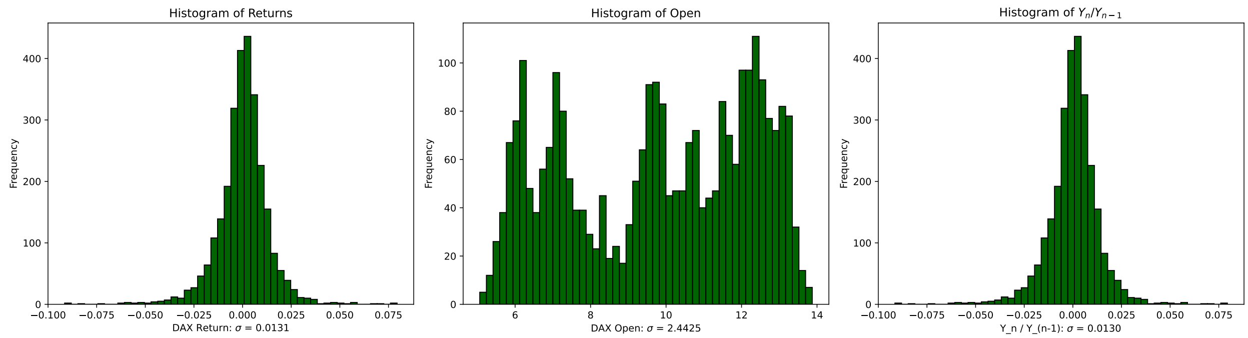


The autocorrelation graph for DAX Returns is below. The correlation of the returns are such that they could be i.i.d.. This means that the relationships between the first price return and its lags aren't as heavily correlated.



1.0.3 Distributions

Now we show the histogram for the opening prices, returns, and volatility. These display the shapes of our data. While returns and volatility are normally distributed, opening prices are more uniformally distributed.



1.0.4 Stationary vs Non Stationary

The time series graphs for DAX Opening prices look nonstationary through visual inspection, and the returns for DAX by day look stationary. We obtain empirical data to show whether the data is stationary or not through the Augmented Dickey Fuller Test. A time series model has a unit root if its first difference is stationary. For a linear time series model, this is equivalent to a $(1 - B)$ factor in the AR polynomial $\phi(B)$. The Augmented Dickey Fuller Test will look for evidence of a unit root. We use the test to verify the DAX Opening prices are nonstationary, and the returns for the DAX stock index are stationary.

$$
\begin{aligned}
H_0 &: \text{The process } \{x_t\} \text{ is nonstationary} \\
&\quad \exists\, t, h, k \text{ such that } (x_t, x_{t+1}, \dots, x_{t+k}) \overset{d}{\neq} (x_{t+h}, x_{t+1+h}, \dots, x_{t+k+h}) \\
H_A &: \text{The process } \{x_t\} \text{ is stationary} \\
&\quad \forall\, t, h, k,\ (x_t, x_{t+1}, \dots, x_{t+k}) \overset{d}{=} (x_{t+h}, x_{t+1+h}, \dots, x_{t+k+h})
\end{aligned}
$$

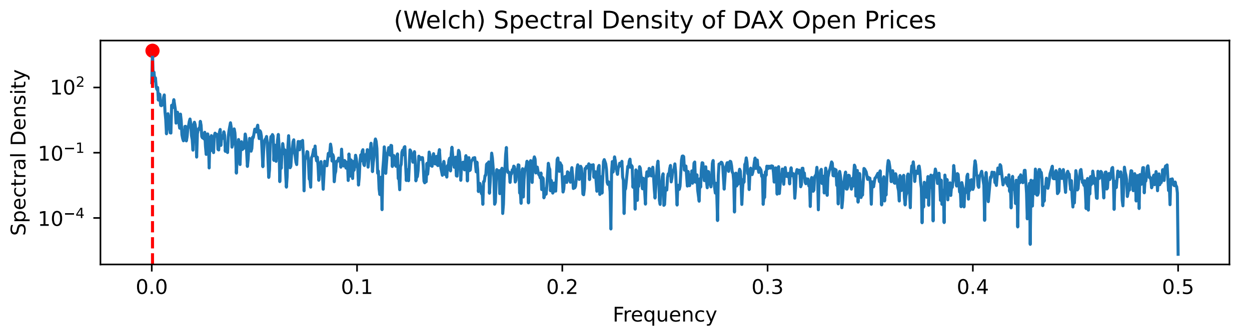
	0	1
Series	DAX Open	DAX Returns
ADF Statistic	-1.35435	-26.565984
p-value	0.603955	0.0
Lags Used	4	3
Observations	2785	2785
Critical Value (5%)	-2.862578	-2.862578
Reject the H0?	No	Yes

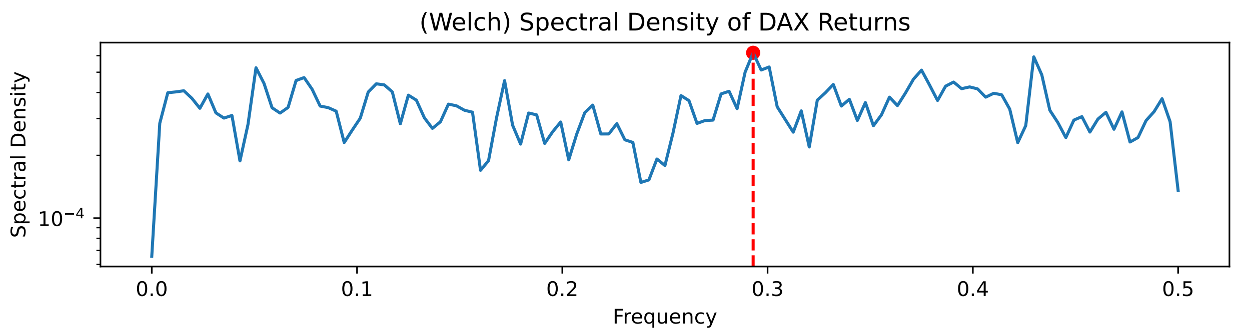
Therefore, the DAX Opens has a p value of 0.603955, so, we fail to reject the null, so we may assume it is nonstationary. On the other hand, the DAX Return has a p value of 0 after running the test. We reject the null and there is evidence to show that the data is stationary.

1.0.5 Spectral Density Function

An estimator of the spectral density function determine a signal's frequency context content from a time series sample. It is measured as $\lambda(\omega) = \int_{\infty}^{-\infty} \gamma(x) e^{-2\pi i \omega x}$ (Ionides 2026). In this section, we look at the frequencies for the DAX Stock Prices and Returns. It splits data into overlapping segments, computes modified periodograms, and averages them to reduce the variance. From the graphs, we may visually inspect the point at which the spectral density is the highest. This will be the dominant frequency, the most common rate at which an oscillation in the stock index happens.

Below, we show frequency vs spectral density function of the DAX Open and Return values.





The highest spectral densities in our graphs above are at frequencies of 0.00036 and 0.3, which indicates that these are the dominant frequencies for DAX open and return values, respectively. The periods are 2777 and 3.3 for the dominant oscillation. These make sense, as, we have roughly 2777 data, so, there is one unique non stationary pattern for open prices. Returns are stationary, and repeat roughly every 3.3 stock measurements, which is roughly 3.3 days.

2 Fitting Time Series Models

Let's train models on time series data to accurately reflect our observations on the DAX Stock Index from the exploratory data analysis. The DAX Open Prices are nonstationary, while the returns are stationary.

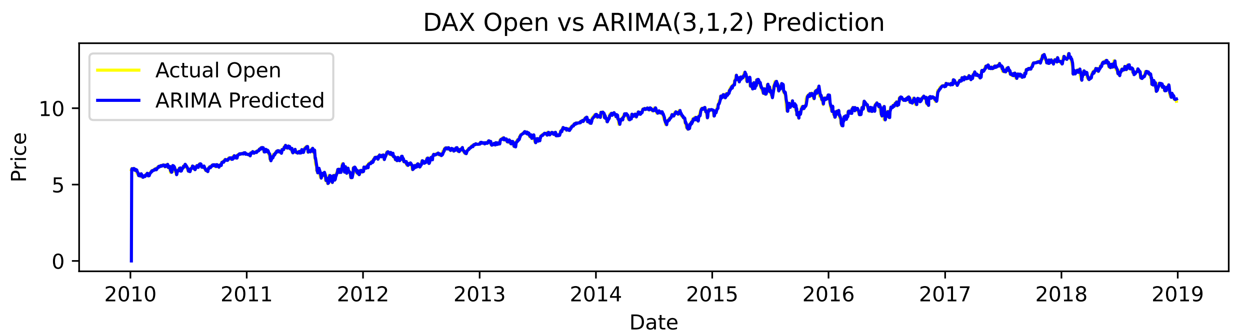
2.0.1 ARMA and ARIMA

How do we forecast DAX Stock Prices and Returns? The exploratory data analysis reveals that these values depend on lags and differencing. This information is captured in an ARMA model.

$$y_n = \phi_1 y_{n-1} + \phi_2 y_{n-2} + \phi_3 y_{n-3} + \theta_1 \epsilon_n + \theta_2 \epsilon_{n-1}$$
$$y_n = \phi_1 B y_n + \phi_2 B^2 y_n + \phi_3 B^3 y_n + \theta_1 \epsilon_n + \theta_2 B \epsilon_n$$
$$y_n \phi(B) = \theta(B) \epsilon_n$$

2.0.1.1 DAX Open Prices

Let's analyze the results of an ARIMA model on the DAX Open Prices; our visual analysis and ADF test revealed that the DAX Open Prices as a time series are nonstationary. Because of the COVID wave in 2020, for purposes of fitting a model, we use all results from Januray 1, 2010 to December 31, 2018. The best ARIMA model for DAX Open Prices is an ARIMA(3,1,2). The shape matches, and has a relatively low AIC of -3916.7 - it is trained and graphed on top of raw data below, overlapping almost perfectly. Before measuring residuals, we compare models by $AIC = 2k - 2ln(\hat{L})$, which punishes complexity and lack of goodness of fit. A lower AIC score tells us that the DAX Stock is modeled better.



The AIC values for the ARIMA models of different number of parameters (ex, ARIMA(1,0,1), ARIMA(2,0,2), etc.) are in the tables below.

Table 2: AIC for ARIMA(p,d,q) Model Comparison

	MA Order				
AR Order	0	1	2	3	4
0	10629.7	7529.3	5005.4	3147.4	1892.3
1	-3897.5	-3895.6	-3893.6	-3893.5	-3892.2
2	-3895.6	-3894.0	-3892.1	-3891.9	-3891.6
3	-3893.6	-3892.4	-3891.1	-3898.6	-3896.7
4	-3893.6	-3891.3	-3889.2	-3896.6	-3895.1

(a) AIC for $d = 0$

	MA Order				
AR Order	0	1	2	3	4
0	-3905.2	-3903.3	-3901.3	-3901.2	-3899.9
1	-3903.3	-3901.3	-3899.3	-3899.3	-3901.8
2	-3901.3	-3899.8	-3916.6	-3906	-3903.5
3	-3901.3	-3899.3	-3916.7	-3904.1	-3903.7
4	-3899.8	-3901.5	-3903.9	-3902.4	-3915.5

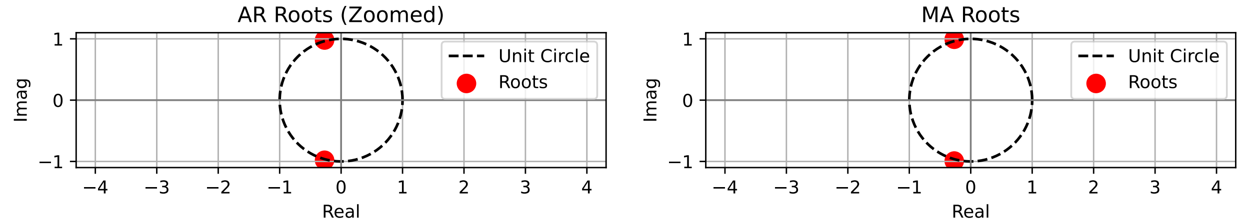
(b) AIC for $d = 1$

	MA Order				
AR Order	0	1	2	3	4
0	-2267.9	-3893.6	-3891.6	-3889.7	-3889.5
1	-2939.6	-3891.6	-3889.6	-3887.7	-3885.7
2	-3253.7	-3889.7	-3887.6	-3887.4	-3904.9
3	-3378.4	-3889.6	-3885.7	-3885.2	-3888.2
4	-3432	-3888.1	-3885.7	-3883.3	-3898.7

(c) AIC for $d = 2$

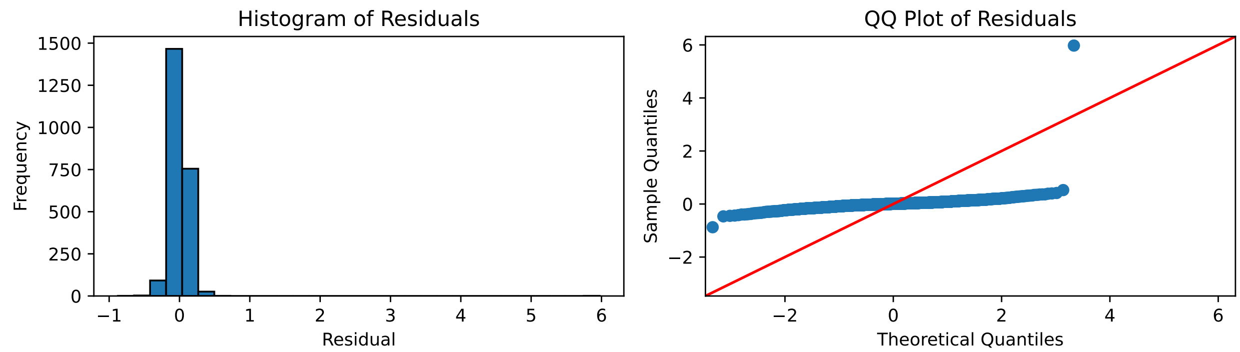
2.0.1.1.1 Model Diagnosis

We check causality and invertibility in this section. Causality is when the AR process has all roots that lie outside the unit circle. Invertability is when the MA process has all roots that lie outside the unit circle. We diagnose our best solution, ARIMA (3,1,2) below. As all roots lie outside the unit circle, our solution is both causal and invertible.



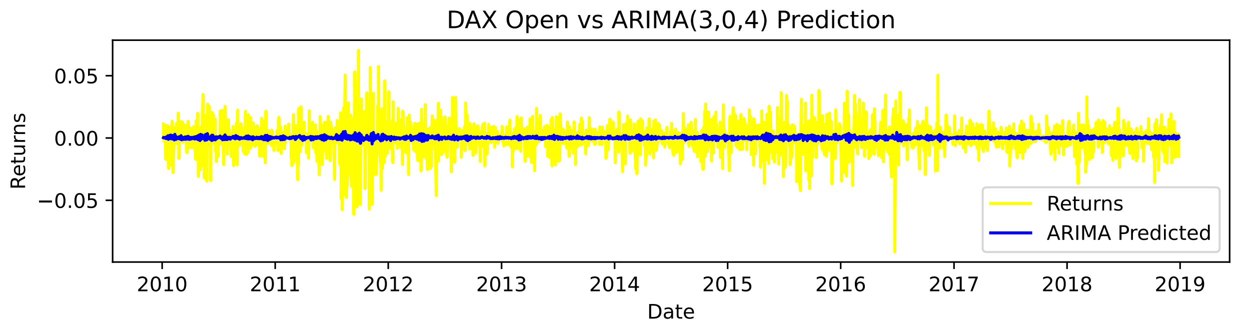
2.0.1.1.2 Residual Analysis

We look at the residuals. Both the qq plot and the histogram indicate that the residuals are normally distributed $\epsilon \approx N(0, \sigma^2)$. This indicates that the ARIMA(3,1,2) is a good fit.



2.0.1.2 DAX Returns

Let's run the same procedure for the returns. The best model is ARIMA(3, 0, 4). The empirical and visual data shows that the returns are stationary, so, having do differencing component ($d = 0$) makes sense for this data.



Now, we look at AIC for every combination of parameters in ARIMA(p,d,q) for DAX returns.

Table 3: AIC for ARIMA(p,d,q) Model Comparison

	MA Order				
AR Order	0	1	2	3	4
0	10629.7	7529.3	5005.4	3147.4	1892.3
1	-3897.5	-3895.6	-3893.6	-3893.5	-3892.2
2	-3895.6	-3894	-3892	-3891.9	-3891.6
3	-3893.6	-3892.4	-3891.1	-3898.6	-3896.7
4	-3893.6	-3891.3	-3889.2	-3896.6	-3895.1

(a) AIC for $d = 0$

	MA Order				
AR Order	0	1	2	3	4
0	-3905.2	-3903.3	-3901.3	-3901.2	-3899.9
1	-3903.3	-3901.3	-3899.3	-3899.3	-3901.8
2	-3901.3	-3899.8	-3916.6	-3906	-3903.5
3	-3901.2	-3899.2	-3916.7	-3904.0	-3903.7
4	-3899.8	-3901.5	-3903.9	-3902.4	-3915.5

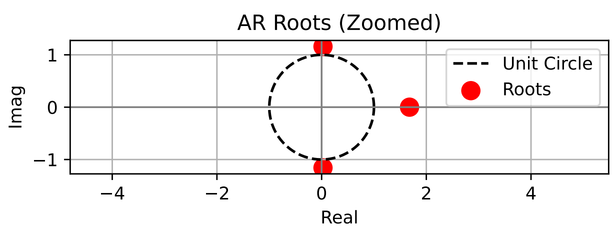
(b) AIC for $d = 1$

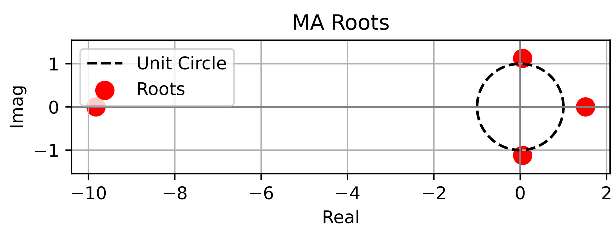
	MA Order				
AR Order	0	1	2	3	4
0	-2267.9	-3893.6	-3891.6	-3889.7	-3889.5
1	-2939.6	-3891.6	-3889.6	-3887.7	-3885.7
2	-3253.7	-3889.7	-3887.6	-3887.4	-3904.9
3	-3378.4	-3889.6	-3885.7	-3885.2	-3888.2
4	-3432	-3888.1	-3885.7	-3883.3	-3898.7

(c) AIC for $d = 2$

2.0.1.2.1 Model Diagnosis

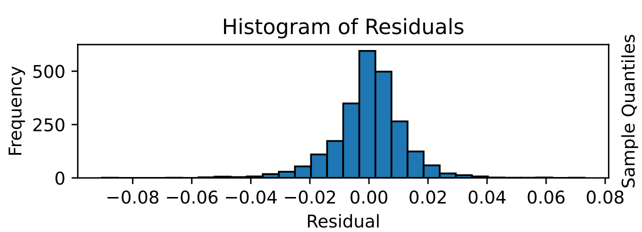
We assess causality and invertability. All solutions lie outside the unit circle. Therefore, our model is causal and invertible.

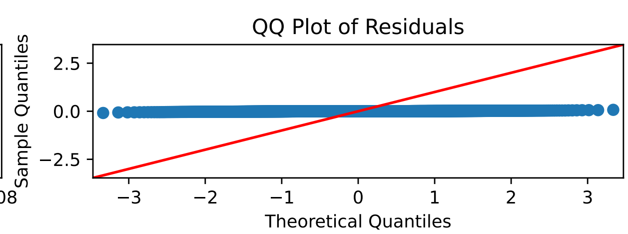




2.0.1.2.2 Residual Analysis

We look at the residuals. Both the qq plot and the histogram indicate that the residuals are normally distributed $\epsilon \approx N(0, \sigma^2)$. This indicates that the ARIMA(3,0,4) is a good fit.





2.0.2 SARMA and SARIMA

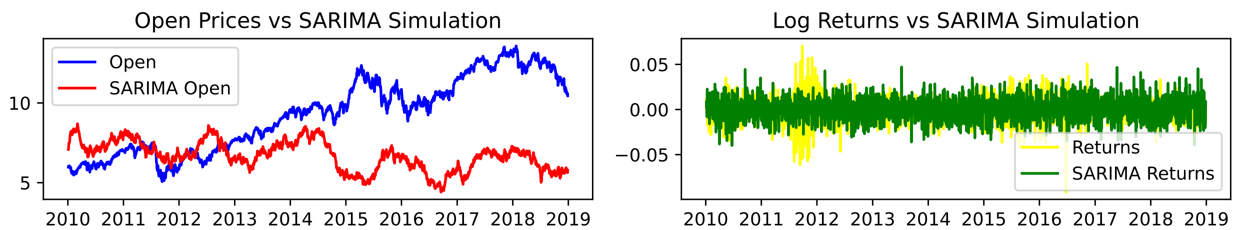
The Seasonal Autoregressive Moving Average Model (SARMA) adds a seasonal component to the ARMA and ARIMA Models. Instead of considering every lag from 1 to 35, for example, we can take lags 1, 12, and 13, repeatedly. This allows for monthly and yearly coefficients in our model, as one would expect with stock. We show a representation of SARIMAX(2,0,2)(1,0,0,12), and how well it performs below.

$$By_n = y_{n-1}; B^2 y_n = y_{n-2}; B^{12} y_n = y_{n-12}$$

$$y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \Phi_1 y_{t-12} + \Phi_1 \phi_1 y_{t-13} + \Phi_1 \phi_2 y_{t-14} = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2}$$

$$(1 - \Phi_1 B^{12})(1 - \phi_1 B - \phi_2 B^2)\, y_t = (1 + \theta_1 B + \theta_2 B^2)\, \epsilon_t$$

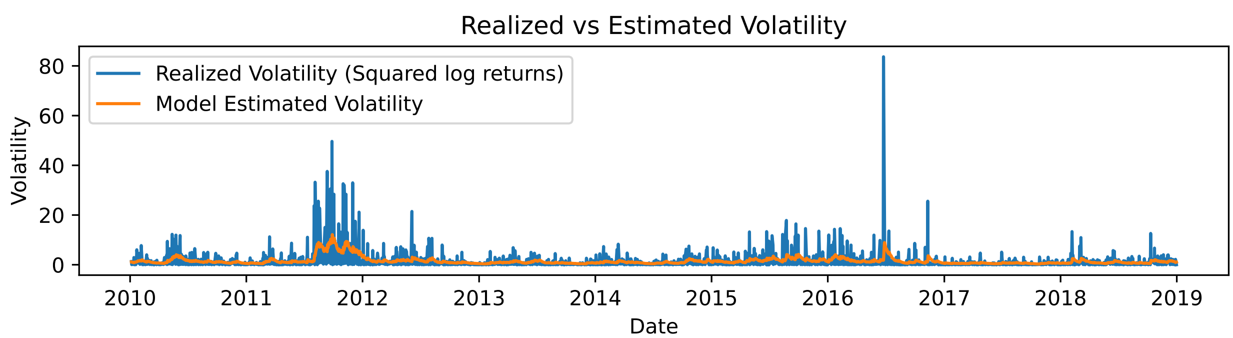
$$\Phi(B^{12})\, \phi(B)\, y_t = \theta(B)\, \Theta(B^{12})\, \epsilon_t$$



The SARIMA models for DAX Open prices and returns have AIC values of -3897.623886161021 and -14029.486910232028.

2.0.3 GARCH

A GARCH model can model volatility of the DAX Stock, shown below.



References

Ionides, Edward L. 2026. *Modeling and Analysis of Time Series Data*. Department of Statistics, University of Michigan. https://ionides.github.io/531w26/01/slides.pdf.

STOXX Ltd. 2026. “DAX® Index.” https://www.stoxx.com/index/dax/.